Vectors I Test

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Full sourced questions. Give exact answers unless otherwise stated.

1 [DHS 2023 Promo Q5]

With reference to the origin O, the points A and B have coordinates $(1, 2, 0)$ and $(-5, 1, 0)$.

(a) Find the angle AOB. [3]

The vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are projected along the direction of $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ to form the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$ respectively.

(b) Find the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$. [3]

(c) Find the exact area of the triangle $O'A'B'$. [2]

2 [NYJC 2024 Promo Q5]

The position vectors of points A, B and C, relative to the origin O are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}, \quad \mathbf{b} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}, \quad \mathbf{c} = 3\mathbf{i} - \mathbf{j} - \mathbf{k}.$$

(a) Find $\mathbf{a} \times \mathbf{b}$ and deduce the exact area of triangle OAB. [3]

(b) Hence, find the volume of the tetrahedron $OABC$.

[The volume of a tetrahedron is $\frac{1}{3} \times$ base area $\times$ perpendicular height.] [3]

(c) Find the perpendicular distance from B to line OA. [2]

3 [MI 2024 Promo Q4]

Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC : AB = 1 : 3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the area of triangle OAC is given by $k|\mathbf{a} \times \mathbf{b}|$, where k is a constant to be determined. [2]

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a} - \mathbf{b}) \cdot \mathbf{c}$. [3]

4 [TMJC 2024 Promo Q6]

(a) The points A, B and C have position vectors $\begin{pmatrix} 1 \\ 3 \\ -7 \end{pmatrix}$, $\begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -9 \\ 8 \\ -27 \end{pmatrix}$, with respect to the origin O.

(i) Show that the points A, B and C are collinear. [2]

(ii) Find the exact area of the triangle OAB. [3]

(b) The vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are defined such that $|\mathbf{u}| = 1$, $|\mathbf{v}| = 2$ and $|\mathbf{w}| = 3$.

(i) Suppose that $\mathbf{u} + \mathbf{v} = \mathbf{w}$. With the aid of a diagram, or otherwise, explain why $\mathbf{u}$ is a unit vector that is parallel to $\mathbf{w}$. [2]

(ii) Given instead that each of the three vectors is perpendicular to the sum of the other two vectors, find the exact value of $|\mathbf{u} + \mathbf{v} + \mathbf{w}|$. [4]

5 [RI 2025 Prelim P1 Q3]

Relative to the origin O, the points A and C have position vectors $\mathbf{a}$ and $2\mathbf{b}$ such that

$$\mathbf{a} = 5p\mathbf{i} - 2p\mathbf{j} + 4p\mathbf{k} \quad \text{and} \quad \mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k},$$

where p is a positive constant. It is given that $|\mathbf{a}| = 2|\mathbf{b}|$.

(a) Find the exact value of p. [2]

(b) Evaluate $(\mathbf{a} + 2\mathbf{b}) \cdot (\mathbf{a} - 2\mathbf{b})$. [1]

(c) Evaluate $\mathbf{a} \times \mathbf{b}$ and hence find the area of triangle OAC. [3]

(d) Use a geometrical reason to explain why $|\mathbf{a} \times \mathbf{b}| = \frac{1}{4}|\mathbf{a} + 2\mathbf{b}||\mathbf{a} - 2\mathbf{b}|$. [2]

END OF TEST